STAT/MA 41600
In-Class Problem Set #35: November 9, 2026
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Problem Set 35 Answers

1a. Let X_i denote the amount of sugar in the ith can. Then we have $P(X_i < 40) = P((X_i - 39)/2 < (40 - 39)/2) = P(Z < 1/2) = 0.6915$, where Z is a standard Normal random variable. Since the amounts of sugar in the cans are independent and identically distributed, it follows that the probability that all 5 cans of soda contain less than 40 grams of sugar each is $(P(X_i < 40))^5 = (0.6915)^5$.

1b. The number of cans of soda that they need to purchase is a Geometric random variable Y with probability mass function $p_Y(y) = (1 - P(X_i \geq 40))^{y-1} P(X_i \geq 40) = (P(X_i < 40))^{y-1}(1 - P(X_i < 40))$ for integers $y \geq 1$, and $p_Y(y) = 0$ otherwise. So they expect to need $E(Y) = 1/(1 - P(X_i < 40)) = 1/(1 - 0.6915) = 3.2415$ cans of soda.

2. The distribution is Binomial with $n = 10$ and $p = P((X - 12)/5 > (15 - 12)/5) = P(Z > 0.6) = 1 - P(Z \leq 0.6) = 1 - 0.7257 = 0.2743$, where we used X to denote the weight of a cat and Z to denote a standard Normal random variable.

3a. Using X to denote the grams of sugar in a can of soda, we want to have $0.25 = P(39 - c < X < 39 + c) = P(X < 39 + c) - P(X \leq 39 - c) = P((X - 39)/2 < c/2) - P((X - 39)/2 \leq -c/2) = P(Z < c/2) - P(Z \leq -c/2)$. So we need $P(Z < c/2) = 0.625$ and $P(Z \leq -c/2) = P(Z \geq c/2) = 1 - P(Z < c/2) = 1 - 0.625 = 0.375$. So we want $0.25 = P(Z < c/2) - P(Z \leq -c/2) = 0.625 - 0.375 = 0.25$. So we use $c/2 = 0.32$ and $c = 0.64$.

3b. Using X to denote the weight of a cat, we want $0.92 = P(12 - c < X < 12 + c) = P(X < 12 + c) - P(X \leq 12 - c) = P((X - 12)/5 < c/5) - P((X - 12)/5 \leq -c/5) = P(Z < c/5) - P(Z \leq -c/5)$. So we need to have $P(Z < c/5) = 0.96$ and $P(Z \leq -c/5) = P(Z \geq c/5) = 1 - P(Z < c/5) = 1 - 0.96 = 0.04$. So we want $0.92 = P(Z < c/5) - P(Z \leq -c/5) = 0.96 - 0.04 = 0.92$. So we use $c/5 = 1.75$ and $c = 8.75$.

4. Let X_i denote the height of the ith blade of grass. Then we have $P(X_i \geq 2) = P((X_i - 4)/0.75 \geq (2 - 4)/0.75) = P(Z \geq -2.67) = P(Z \leq 2.67) = 0.9962$, where Z is a standard Normal random variable. Since the heights of the 100 blades of grass are independent and identically distributed, it follows that the probability that all 100 blades of grass have height 2 inches or more is $(P(X_i \geq 2))^{100} = (0.9962)^{100} = 0.68$.

5a. Let X be the sugar content in a can of soda. We get $P(X > 42.6 \mid X > 40.8) = P(X > 42.6)/P(X > 40.8) = P((X - 39)/2 > (42.6 - 39)/2)/P((X - 39)/2 > (40.8 - 39)/2) = P(Z > 1.8)/P(Z > 0.9) = (1 - P(Z \leq 1.8))/(1 - P(Z \leq 0.9)) = (1 - 0.9641)/(1 - 0.8159) = (0.0359)/(0.1841) = 0.1950$, where Z is a standard Normal random variable.

5b. Let X be the weight of a cat. We get $P(X > 14 \mid X > 13) = P(X > 14)/P(X > 13) = P((X-12)/5 > (14-12)/5)/P((X-12)/5 > (13-12)/5) = P(Z > 0.4)/P(Z > 0.2) = (1-P(Z \leq 0.4))/(1-P(Z \leq 0.2)) = (1-0.6554)/(1-0.5793) = (0.3446)/(0.4207) = 0.8191$, where Z is a standard Normal random variable.

5c. Let X be the height of a blade of grass. We get $P(X < 3.2 \mid X < 3.8) = P(X < 3.2)/P(X < 3.8) = P((X - 4)/0.75 < (3.2 - 4)/0.75)/P((X - 4)/0.75 < (3.8 - 4)/0.75) = P(Z < -1.07)/P(Z < -0.27) = P(Z > 1.07)/P(Z > 0.27) = (1-P(Z \leq 1.07))/(1-P(Z \leq 0.27)) = (1 - 0.8577)/(1 - 0.6064) = (0.1423)/(0.3936) = 0.3615$, where Z is a standard Normal random variable.